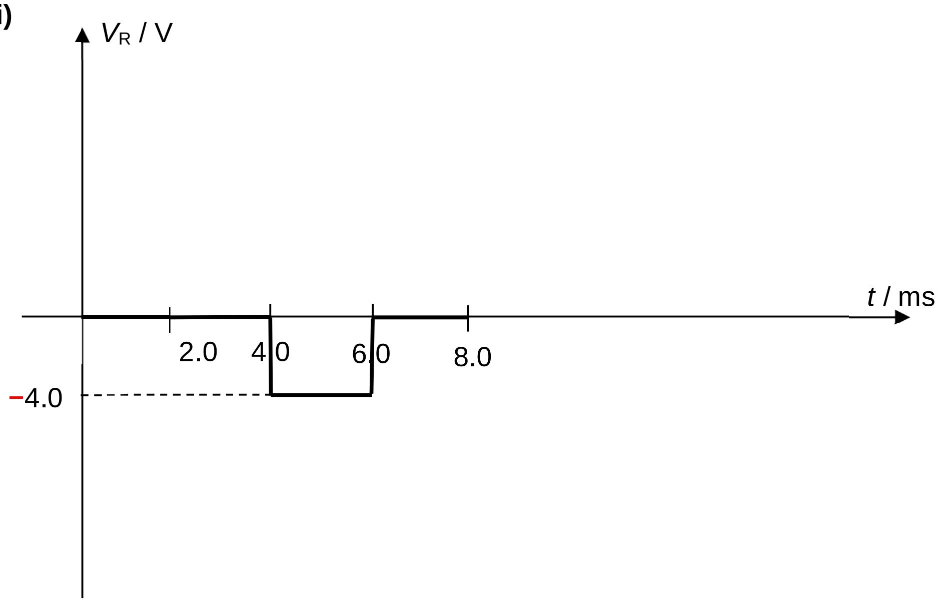
6	**(a)**	**(i)**	$V_{rms} = \sqrt{\dfrac{2(5.0)^2(2.0)+2(10.0)^2(2.0)}{8.0}}$ [M1] $= 7.9$ V [A1]	
		(ii) 1.	0 V	[A1]
		(ii) 2.	p.d. across resistor R = − 10.0 + 6.0 = − 4.0 V OR + 4.0 V	[M1] [A1]

(iii)



1 mark for shape of graph.
1 mark for labelled axes (values of V_R if graph is correct)
Allow + 4.0 V

	(b)	Transmission voltage can be stepped up [B1], thereby minimising power loss [B1].

7	(a)	$^{226}\text{Ra} \rightarrow {}^{222}\text{Rn} + {}^{4}\text{He}$	